A FILTRATION BOUND FOR THE ERASURE ENTROPIES OF BOOLEAN FUNCTIONS

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ABSTRACT. For a Boolean function f and a coordinate i, the erasure entropy $E_i(f)$ is the conditional entropy of the event $i \in U$ given $U \setminus \{i\}$, where U is drawn from the spectral distribution $\hat{f}^2$. The Fourier entropy-influence conjecture is equivalent to the existence of an absolute c such that every non-constant f has a coordinate with $E_i(f) \le c\,\mathrm{Inf}_i(f)$. We bound E_i from above by revealing the other coordinates one at a time and conditioning only on the last revealed coordinate that the spectral sample contains. The resulting quantity L_i^π depends on the order π, is a sum, over the steps, of a two-variable entropy of the squared increments of the Doob martingales of the average of f over x_i and of its derivative $D_i f$, lies between E_i and the binary entropy of Inf_i, and can be minimised over all orders by a shortest-path recursion. We show that $L_i^\pi = O(\mathrm{Inf}_i)$ whenever, along the order, the probability that i is pivotal both before and after the not yet revealed coordinates are resampled, times the block influence of those coordinates, is $O(\mathrm{Inf}_i^2)$. We also give an exact formula for E_i under block composition, and show that a coordinate whose variable enters only through one AND of $d+1$ variables, pivotal with probability ρ, has $L_i^\pi/\mathrm{Inf}_i \le 2 + 2^{-d}(2d + \log_2(1/\rho))$ for a suitable order. Conditioning on the last r revealed coordinates gives a decreasing family of bounds that reaches E_i at $r = n-1$.

1. INTRODUCTION

For $f\colon \{-1,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$, and let U be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$ and the total influence is $I(f) = \mathbb{E}|U| = \sum_i \mathrm{Inf}_i(f)$, where $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [1] asks for an absolute C with $H(f) \le C\,I(f)$ for every Boolean f. It is known for symmetric functions and read-once decision trees [9] and for linear threshold functions [13], and it is preserved under suitable compositions [8]. There are bounds in terms of decision-tree and communication complexity measures [3, 15]; Arunachalam, Chakraborty, Koucký, Saurabh and de Wolf [2] bound $H(f)$ by twice the average unambiguous parity-certificate complexity. Kelman, Kindler, Lifshitz, Minzer and Safra [6] prove a weak form for the min-entropy. For every Boolean f, Han [5] proved

$$H(f) \le \Big(2 + \frac{3}{\ln 2}\Big) I(f) + \sum_i \mathrm{Inf}_i(f) \log_2 \frac{1}{\mathrm{Inf}_i(f)};$$

Li and Han [7] lowered the constant in front of $I(f)$ to the sharp value $1/\ln 2$, and a form with $\mathrm{Var}(f)/\mathrm{Inf}_i(f)$ in place of $1/\mathrm{Inf}_i(f)$ in the logarithm is in [12].

One route to the conjecture goes through a single coordinate at a time. The drop of the entropy from f to its two restrictions at $x_i = \pm 1$ splits into the erasure entropy (in the sense of Verdú and Weissman [14])

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big)$$

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where $\mathbf{1}[\cdot]$ is the indicator, and a Jensen–Shannon term $J_i(f) \le (2/\ln 2)\mathrm{Inf}_i(f)$. The route goes back to Han [5, Remarks 3.1–3.2]; the splitting is equation (32) of González, MacManus and Pereyra [4]. In [10] it is shown that the conjecture holds if and only if there is an absolute c such that every non-constant Boolean function has a coordinate i with

$$E_i(f) \le c\,\mathrm{Inf}_i(f), \tag{1}$$

and in [11] this quantity is studied directly. The difficulty is that E_i conditions on the whole of $U \setminus \{i\}$, an object with up to 2^{n-1} values. The trivial bound $E_i \le H(\mathbf{1}[i \in U])$ loses a factor $\log(1/\mathrm{Inf}_i)$, and (1) asks exactly to remove that factor at one coordinate.

Here we condition on less. Reveal the other coordinates in a fixed order and record only the position of the last revealed coordinate that U contains. This gives an upper bound L_i^π for E_i (Theorem 1) with a martingale form: at each step it compares how much of the average of f over x_i and how much of the derivative $D_i f$ the newly revealed coordinate explains. A two-sequence inequality turns a product condition along the order into the bound $L_i^\pi = O(\mathrm{Inf}_i)$ (Theorem 4), and the product condition has a direct reading in terms of resampling. Section 4 gives an exact composition formula for E_i, Section 5 the case of a private AND, and Section 6 the windows of length r, which record the last r revealed coordinates that U contains. Section 7 lists exact values.

2. The witness

Fix i and write g for the average of f over x_i and $h = D_i f$ for its derivative, functions of the other coordinates

$$g = \tfrac{1}{2}\big(f|_{x_i=1} + f|_{x_i=-1}\big), \qquad h = \tfrac{1}{2}\big(f|_{x_i=1} - f|_{x_i=-1}\big).$$

They take values in $\{-1, 0, 1\}$, have disjoint supports, $\hat{g}(T) = \hat{f}(T)$ and $\hat{h}(T) = \hat{f}(T \cup \{i\})$ for $T \not\ni i$, $\mathbb{E}h^2 = \mathrm{Inf}_i(f) =: \delta$ and $\mathbb{E}g^2 = 1 - \delta$. Put $p(T) = \hat{f}(T)^2$ and $q(T) = \hat{f}(T \cup \{i\})^2$ for $T \subseteq [n] \setminus \{i\}$, and

$$\Phi(a, b) = \eta(a) + \eta(b) - \eta(a + b) = (a + b)\, h_2\Big(\frac{b}{a + b}\Big), \qquad \eta(x) = -x \log_2 x,$$

where h_2 is the binary entropy. Then $E_i(f) = \sum_T \Phi(p(T), q(T))$, since $(U \setminus \{i\}, \mathbf{1}[i \in U])$ takes the value $(T, 0)$ with probability $p(T)$ and $(T, 1)$ with probability $q(T)$. The function Φ is concave, homogeneous of degree one and increasing in each argument; in particular it is superadditive.

Let $\pi = (j_1, \dots, j_m)$, $m = n-1$, be an order of the other coordinates, and let $S_k = \{j_1, \dots, j_k\}$. For $T \subseteq [n] \setminus \{i\}$ let $\mathrm{lab}(T) = \max\{k : j_k \in T\}$, with $\mathrm{lab}(\emptyset) = 0$, and put

$$A_k = \sum_{\mathrm{lab}(T)=k} p(T), \qquad B_k = \sum_{\mathrm{lab}(T)=k} q(T), \qquad L_i^\pi(f) = \sum_{k=0}^{m} \Phi(A_k, B_k).$$

Theorem 1. *For every Boolean f, coordinate i and order π:*

(a) $E_i(f) \le L_i^\pi(f) \le h_2(\mathrm{Inf}_i(f))$.

(b) *With $\mathcal{F}_k$ the σ-algebra generated by $x_{j_1}, \dots, x_{j_k}$,*

$$A_k = \mathbb{E}\big(\mathbb{E}[g \mid \mathcal{F}_k] - \mathbb{E}[g \mid \mathcal{F}_{k-1}]\big)^2, \qquad B_k = \mathbb{E}\big(\mathbb{E}[h \mid \mathcal{F}_k] - \mathbb{E}[h \mid \mathcal{F}_{k-1}]\big)^2$$

for $k \ge 1$, and $A_0 = (\mathbb{E}g)^2$, $B_0 = (\mathbb{E}h)^2$.

(c) *Let $a(R) = \sum_{T \subseteq R} p(T)$ and $b(R) = \sum_{T \subseteq R} q(T)$. Then $\min_\pi L_i^\pi = \lambda(R_*)$ for the full set $R_* = [n] \setminus \{i\}$, where $\lambda(\emptyset) = \Phi(a(\emptyset), b(\emptyset))$ and*

$$\lambda(R) = \min_{j \in R}\Big(\lambda(R \setminus \{j\}) + \Phi\big(a(R) - a(R \setminus \{j\}),\, b(R) - b(R \setminus \{j\})\big)\Big).$$

Proof. (a) The pair $(\mathrm{lab}(U\setminus\{i\}), \mathbf{1}[i\in U])$ takes the value $(k,0)$ with probability A_k and $(k,1)$ with probability B_k, so $L_i^\pi = H(\mathbf{1}[i\in U] \mid \mathrm{lab}(U\setminus\{i\}))$. Since $\mathrm{lab}(U\setminus\{i\})$ is a function of $U\setminus\{i\}$, conditioning on it cannot give a smaller entropy than conditioning on $U\setminus\{i\}$, and it cannot give a larger one than conditioning on nothing, which is $h_2(\delta)$.

(b) $\mathbb{E}[g \mid \mathcal{F}_k] = \sum_{T\subseteq S_k} \hat g(T)\chi_T$, so the k-th increment is $\sum_{\mathrm{lab}(T)=k} \hat g(T)\chi_T$, whose squared norm is A_k by Parseval. The same holds for h.

(c) The sets $\{T : \mathrm{lab}(T)\le k\}$ are the subsets of S_k, so $A_k = a(S_k) - a(S_{k-1})$ and $B_k = b(S_k) - b(S_{k-1})$. An order is a maximal chain $\emptyset = S_0 \subset S_1 \subset \cdots \subset S_m = R_*$, and L_i^π is a sum of costs along its edges. The recursion is the shortest path in this graph. □

The recursion costs $O(m2^m)$ operations per coordinate, against $m!$ orders. Part (b) says that L_i^π follows the Doob martingales of g and of h along the order and charges, at each step, the entropy of telling the two increments apart. A coordinate is good along π when h is explained late, after most of g has been explained.

3. A sufficient condition

Lemma 2. *Let $A_k, B_k \ge 0$ for $k = 0,\dots,m$, let $\delta = \sum_k B_k > 0$, and suppose that for some $M \ge 0$*

$$\Big(\sum_{k\ge K} A_k\Big)\Big(\sum_{k\le K} B_k\Big) \le M\delta^2 \qquad \text{for every } K = 0,\dots,m.$$

Then $\sum_k \Phi(A_k,B_k) \le \big(1/\ln 2 + 4 + 2\log_2(1+2M)\big)\delta$.

Proof. Since $a\log_2(1+b/a) \le b/\ln 2$, we have $\Phi(a,b) \le b/\ln 2 + \psi(a,b)$ with $\psi(a,b) = b\log_2(1+a/b)$. The function ψ is concave, homogeneous of degree one and increasing in each argument, so $\sum_{k\in G}\psi(A_k,B_k) \le \psi(A_G,B_G)$ for every set G of indices, where $A_G = \sum_{k\in G} A_k$. Put $\beta_k = \delta^{-1}\sum_{l\le k} B_l$, which is nondecreasing, and $G_t = \{k : 2^{-t-1} < \beta_k \le 2^{-t}\}$ for $t \ge 0$. Indices with $\beta_k = 0$ have $B_k = 0$ and contribute nothing. For $k_t = \min G_t$ the hypothesis gives $A_{G_t} \le \sum_{k\ge k_t} A_k \le M\delta/\beta_{k_t} \le 2^{t+1}M\delta$, and $B_{G_t} \le \beta_{\max G_t}\delta \le 2^{-t}\delta$. Hence

$$\sum_{k\in G_t} \psi(A_k,B_k) \le 2^{-t}\delta\log_2\big(1 + 2^{2t+1}M\big) \le 2^{-t}\delta\big(2t + \log_2(1+2M)\big),$$

and summing over t with $\sum_t 2^{-t} = 2$ and $\sum_t t2^{-t} = 2$ gives the claim. □

Remark 3. Under the hypothesis of Lemma 2 one also has $\sum_k \Phi(A_k,B_k) \le \big(4/\ln 2 + \log_2(1+2M)\big)\delta$, which is the smaller bound when $M \ge 0.13$. Work in natural logarithms, put $\kappa = 1+2M$ and let β_k be as in the proof of Lemma 2; indices with $B_k = 0$ contribute only $A_k\ln 1 = 0$. First, $a\ln(1+b/a) \le b$. Second, for $B_k > 0$ we have $\beta_k > 0$, and $\ln x \le x-1$ at $x = \beta_k^2(A_k+B_k)/(\kappa B_k)$ gives

$$B_k\ln\Big(1 + \frac{A_k}{B_k}\Big) \le B_k\ln\kappa - 2B_k\ln\beta_k + \frac{\beta_k^2 A_k}{\kappa},$$

since $\beta_k^2 \le 1 \le \kappa$. Third, $\sum_k B_k(-\ln\beta_k) = \delta\sum_k(\beta_k - \beta_{k-1})(-\ln\beta_k) \le \delta\int_0^1(-\ln x)\,dx = \delta$, with $\beta_{-1} = 0$. Fourth, with $\tau_l = \sum_{k\ge l} A_k \le M\delta/\beta_l$ (terms with $\beta_l = 0$ vanish),

$$\sum_k A_k\beta_k^2 = \sum_l(\beta_l^2 - \beta_{l-1}^2)\tau_l \le M\delta\sum_l \frac{(\beta_l - \beta_{l-1})(\beta_l + \beta_{l-1})}{\beta_l} \le 2M\delta.$$

Adding, $\sum_k \Phi(A_k,B_k)\ln 2 \le \delta + \delta\ln\kappa + 2\delta + 2M\delta/\kappa \le (4 + \ln\kappa)\delta$. In particular the constant in Theorem 4 may be replaced by $4/\ln 2 + \log_2(1 + 2\max(M,1))$.

For a set R of coordinates other than i, let x^R be x with the coordinates in R replaced by independent uniform bits, and put

$$W(R) = \mathbb{E}\, h(x)h(x^R) = \Pr[i \in U,\ U \cap R = \emptyset],$$
$$V(R) = \mathbb{E}g^2 - \mathbb{E}\, g(x)g(x^R) = \Pr[i \notin U,\ U \cap R \neq \emptyset].$$

$W(R)$ is the correlation of the pivotality of i, with its sign, at x and at x^R, and $V(R)$ is the influence of the block R on g. The two equalities are Parseval: $\mathbb{E}\,\varphi(x)\varphi(x^R) = \sum_{T\cap R=\emptyset} \hat{\varphi}(T)^2$ for every real function φ of the coordinates other than i.

Theorem 4. *Let π be an order and $R_k = R_* \setminus S_k$ the set of coordinates not yet revealed after step k. Then $\sum_{l\le k} B_l = W(R_k)$ and $\sum_{l\ge k} A_l = V(R_{k-1})$. Consequently, if*

$$W(R_k)\, V(R_{k-1}) \le M \operatorname{Inf}_i(f)^2 \qquad \text{for } k = 1, \dots, m,$$

then $L_i^\pi(f) \le \big(1/\ln 2 + 4 + 2\log_2(1 + 2\max(M,1))\big)\operatorname{Inf}_i(f)$.

Proof. $\operatorname{lab}(T) \le k$ means $T \subseteq S_k$, that is $T \cap R_k = \emptyset$, which gives the first identity; $\operatorname{lab}(T) \ge k$ means $T \cap R_{k-1} \neq \emptyset$, which gives the second. For $K = 0$ the product in Lemma 2 is $(1 - \delta)\hat{f}(\{i\})^2 \le \delta^2$, because $|\hat{f}(\{i\})| = |\mathbb{E}h| \le \Pr[h \neq 0] = \delta$. Apply Lemma 2 with $\max(M, 1)$. □

In words: i is good along π if at every stage either the value of g is almost settled by the revealed coordinates (the resampled block has small influence on g), or the revealed coordinates say little about whether i is pivotal (resampling the rest destroys the pivotality of i). Because W decreases and V increases with R, a coarser chain $R_* = R'_0 \supset R'_1 \supset \cdots \supset \emptyset$ that the order refines suffices: if $W(R'_{l+1})V(R'_l) \le M\operatorname{Inf}_i(f)^2$ for every l, the condition of Theorem 4 holds for every order that reveals $R'_l \setminus R'_{l+1}$ in block l.

Write $\mathcal{D} = \{h \neq 0\}$, $\mathcal{L} = \{g = -1\}$ and $\mathcal{H} = \{g = 1\}$. Since $g \in \{-1, 0, 1\}$ and $V(R) = \frac{1}{2}\mathbb{E}(g(x) - g(x^R))^2$, every Boolean f has $V(R) = \Pr[x \in \mathcal{D},\, x^R \notin \mathcal{D}] + 4\Pr[x \in \mathcal{L},\, x^R \in \mathcal{H}]$; if f is monotone, then $h \ge 0$ and $W(R) = \Pr[x \in \mathcal{D},\, x^R \in \mathcal{D}]$. For monotone f the condition reads: the probability of staying in $\mathcal{D}$ times the probability of jumping across it is $O(\Pr[\mathcal{D}]^2)$.

For example, let f be the OR of s disjoint ANDs of width w, let i lie in the first AND, let Q be the set of the other $w-1$ variables of that AND, and let $P_0 = (1-2^{-w})^{s-1}$ be the probability that no other AND is satisfied. Take the chain $R_* \supset Q \supset \emptyset$, that is, reveal the other ANDs first and Q last. Here $x \in \mathcal{D}$ when no other AND is satisfied and every variable of Q is true, so $\delta = \operatorname{Inf}_i(f) = P_0 2^{1-w}$ and $W(Q) = P_0 4^{1-w} = \delta^2/P_0$; with $V(R_*) \le 1$ the first product is at most δ^2/P_0. A move from $\mathcal{L}$ to $\mathcal{H}$ needs another AND to change, so resampling Q cannot make it, and $V(Q) \le \Pr[x \in \mathcal{D}] = \delta$; with $W(\emptyset) = \delta$ the second product is at most δ^2. The condition therefore holds with $M = 1/P_0$, which is bounded in the usual regime $s \approx 2^w \ln 2$, where P_0 is close to $\frac{1}{2}$.

Conjecture 5 (filtration form)**.** *There is an absolute c such that every non-constant Boolean f has a coordinate i and an order π of the other coordinates with $L_i^\pi(f) \le c\operatorname{Inf}_i(f)$.*

By Theorem 1(a) and [10], Conjecture 5 implies the FEI conjecture with constant $c + 2/\ln 2$: the class of Boolean functions is closed under restriction, and the one-coordinate step applies at the coordinate given by the conjecture in every restriction. Conjecture 5 is a strengthening of (1); Section 7 shows no counterexample among the functions we can evaluate exactly, and Section 6 discusses where a single last element may not be enough.

4. Composition

Proposition 6. *Let X and Z be disjoint sets of coordinates, let $\gamma\colon \{-1,1\}^X \to \{-1,1\}$ and $F\colon \{-1,1\} \times \{-1,1\}^Z \to \{-1,1\}$ be Boolean, and put $f(x_X, z) = F(\gamma(x_X), z)$. Write $F(u, z) =$*

$F_0(z) + uF_1(z)$. *For* $i \in X$,

$$\mathrm{Inf}_i(f) = \mathrm{Inf}_i(\gamma)\,\|F_1\|_2^2,$$

$$E_i(f) = \|F_1\|_2^2\Big(E_i(\gamma) - \Phi\big(\hat\gamma(\emptyset)^2, \hat\gamma(\{i\})^2\big)\Big) + \sum_{T\subseteq Z} \Phi\Big(\big(\hat F_0(T) + \hat\gamma(\emptyset)\hat F_1(T)\big)^2,\ \hat\gamma(\{i\})^2\hat F_1(T)^2\Big).$$

Proof. For $T_1 \subseteq X$ and $T_2 \subseteq Z$, $\hat f(T_1 \cup T_2) = [T_1 = \emptyset]\hat F_0(T_2) + \hat\gamma(T_1)\hat F_1(T_2)$. For $i \notin T_1 \neq \emptyset$ the pair (p, q) at $T_1 \cup T_2$ is $\hat F_1(T_2)^2$ times the pair of γ at T_1, so by homogeneity these terms sum to $\|F_1\|^2$ times the terms of $E_i(\gamma)$ with $T_1 \neq \emptyset$. The terms with $T_1 = \emptyset$ form the last sum. The influence is the sum of q. □

The last sum depends on γ only through $\hat\gamma(\emptyset)$ and $\hat\gamma(\{i\})$. When γ is itself a composition this can be iterated, which gives E_i and Inf_i of iterated block compositions from the spectra of the gates; the iteration for one gate applied to copies of another, carrying four numbers per leaf through each level, is in [11, Section 2], where it is used for towers of transitive gates. O'Donnell and Tan [8] prove a composition theorem for the conjecture itself.

5. A private AND

Proposition 7. *Let* i *and a set* Y *of* $d \geq 0$ *other coordinates enter* f *only through* $\omega = \mathbf{1}[x_i = x_j = 1 \text{ for all } j \in Y]$, *so that* $f = f_0(v) + f_1(v)\,\omega$ *with* v *the remaining coordinates and* $f_1 \in \{0, \pm 2\}$. *Let* $\delta = \mathrm{Inf}_i(f) = 2^{-d}\Pr[f_1 \neq 0] > 0$ *and let* π *reveal* v *first and* Y *last. Then*

$$2\big(1 - 2^{-d}\big) \leq \frac{E_i(f)}{\delta} \leq \frac{L_i^\pi(f)}{\delta} \leq 2\big(1 - 2^{-d}\big) + 2^{-d}\log_2 \frac{e\,2^d}{\delta}.$$

Proof. In ± 1 notation $\omega = 2^{-(d+1)}\sum_{S\subseteq Y\cup\{i\}} \chi_S$. Hence for $T_0 \subseteq v$ and $S \subseteq Y$, the coefficient of f at $T_0 \cup S \cup \{i\}$ is $2^{-(d+1)}\hat f_1(T_0)$, and at $T_0 \cup S$ it is the same number when $S \neq \emptyset$. Put $p_{T_0} = 4^{-(d+1)}\hat f_1(T_0)^2$; then $\delta = \sum_S q = 2^d \sum_{T_0} p_{T_0}$. Every $T = T_0 \cup S$ with $S \neq \emptyset$ has $p(T) = q(T) = p_{T_0}$ and contributes $\Phi(p_{T_0}, p_{T_0}) = 2p_{T_0}$; there are $2^d - 1$ such S for each T_0, which gives $2(1 - 2^{-d})\delta$, and the remaining terms of E_i are nonnegative. In the witness, the label of $T_0 \cup S$ with $S \neq \emptyset$ is the position of the last element of S; the l-th element of Y is the last element of 2^{l-1} sets S, so the atom at the position of that element has $A_k = B_k = 2^{l-1}\sum_{T_0} p_{T_0}$ and contributes $2^l \sum_{T_0} p_{T_0}$, again $2(1-2^{-d})\delta$ in total. The atoms of the sets $T = T_0$ carry q-mass $\nu = 2^{-d}\delta$ and p-mass $\mu = 1 - 2\delta + 2^{-d}\delta$, so $\mu + \nu \leq 1$. By superadditivity their contribution is at most $\Phi(\mu, \nu) \leq \nu/\ln 2 + \nu\log_2((\mu+\nu)/\nu) \leq \nu\log_2(e/\nu)$. □

Since $\delta = 2^{-d}\Pr[f_1 \neq 0]$, the right-hand side is $2 + 2^{-d}\big(2d + \log_2 e - 2 + \log_2(1/\Pr[f_1 \neq 0])\big)$, which is less than $2 + 2^{-d}\big(2d + \log_2(1/\Pr[f_1 \neq 0])\big)$, so the ratio tends to 2 as the private AND grows, unless it is pivotal with probability below about 2^{-2^d}. The same computation applies to any private gadget whose Fourier coefficients at S and $S \cup \{i\}$ agree in absolute value for every non-empty S. Width-dependent bounds for every DNF, with no privacy assumption, are in [11].

6. Windows

For $1 \leq r \leq m$ let $\mathrm{lab}_r(T)$ be the set of the last r positions of π met by T (all of them if $|T| < r$), and

$$L_i^{\pi,r}(f) = H\big(\mathbf{1}[i \in U] \,\big|\, \mathrm{lab}_r(U \setminus \{i\})\big).$$

Then $L_i^{\pi,1} = L_i^\pi$, $L_i^{\pi,m} = E_i$, and

$$E_i(f) \leq L_i^{\pi,r}(f) \leq L_i^{\pi,r-1}(f),$$

because lab_{r-1} is a function of lab_r, which is a function of $U \setminus \{i\}$. The masses of the atoms of $L_i^{\pi,r}$ are signed sums, by inclusion and exclusion, of resampling correlations of derivatives of order at most r: for a set $\mathcal{K}$ of coordinates other than i and a set R of coordinates disjoint from it, $\Pr[i \in U, \mathcal{K} \subseteq U, U \cap R = \emptyset] = \mathbb{E}\, D_{\mathcal{K}} h(x)\, D_{\mathcal{K}} h(x^R)$, where $D_{\mathcal{K}}$ is the product of the derivatives D_j over $j \in \mathcal{K}$.

The ladder runs from Conjecture 5 at $r = 1$ to (1) at $r = m$. Window 1 sees only the last element of $U \setminus \{i\}$. When every variable lies in many overlapping certificates, the last element alone may carry little information about whether $i \in U$, while a pair of elements from one certificate of i may carry much more. We do not know whether Conjecture 5 holds, or whether it holds with some fixed window r in place of 1.

7. Exact values

The statements of Theorems 1 and 4, Lemma 2 and Propositions 6, 7 were checked with exact Walsh–Hadamard transforms. On 58 random non-constant functions of 3 to 7 variables (each value -1 with probability $\frac{1}{2}$, 0.2 or 0.05), at every coordinate and for every order when $n \le 6$ and 150 random orders when $n = 7$: the bounds of Theorem 1(a) hold to $3 \cdot 10^{-16}$, the shortest path of Theorem 1(c) agrees with the minimum over all orders to $3 \cdot 10^{-16}$, the martingale form and the resampling identities hold exactly, the bound of Lemma 2 holds with the smallest admissible M, and $E_i \le L_i^{\pi,3} \le L_i^{\pi,2} \le L_i^{\pi,1}$ with $L_i^{\pi,m} = E_i$. The composition formula on 60 random compositions to $4 \cdot 10^{-16}$, and the bracket of Proposition 7 on 80 random private ANDs. The table gives $\min_{i,\pi} L_i^\pi/\mathrm{Inf}_i$ and $\min_i E_i/\mathrm{Inf}_i$; $\mathrm{TRIBES}_{w\times s}$ is the OR of s disjoint ANDs of width w.

	AND_6	AND_8	AND_{10}	$\mathrm{TRIBES}_{2\times 3}$	$\mathrm{TRIBES}_{2\times 4}$	$\mathrm{TRIBES}_{2\times 5}$
$\min_{i,\pi} L_i^\pi/\mathrm{Inf}_i$	2.2922	2.1048	2.0341	2.3160	3.0428	3.3892
$\min_i E_i/\mathrm{Inf}_i$	2.2922	2.1048	2.0341	2.3160	3.0428	3.3892
	MAJ_3	MAJ_5	MAJ_7	MAJ_9	MAJ_{11}	$\mathrm{TRIBES}_{3\times 3}$
$\min_{i,\pi} L_i^\pi/\mathrm{Inf}_i$	1.0000	1.5204	1.7848	1.9578	2.0844	2.5474
$\min_i E_i/\mathrm{Inf}_i$	0	0	0	0	0	2.5474

On AND and tribes the witness of the best order equals E_i at the best coordinate. The functions AND_n are the case $v = \emptyset$ of Proposition 7, and they sit just below its upper bound: at $n = 10$, $\min_{i,\pi} L_i^\pi/\mathrm{Inf}_i = 2.034057$ against 2.034068. On majority $E_i = 0$, because the spectrum lives on odd sets and $U \setminus \{i\}$ determines the parity of $|U|$, while the witness pays for the last element; its growth along n is slow and stays below 2.1 at $n = 11$.

The coordinate matters. For even m let $\mathrm{IP}(y) = (-1)^{N(y)}$ on $\{-1,1\}^m$, where $N(y)$ is the number of $j \le m/2$ with $y_j = y_{j+m/2} = -1$ (the inner product function), and let $f(y, x_i) = \mathrm{IP}(y)$ for $y \ne (1,\dots,1)$ and $f((1,\dots,1), x_i) = x_i$. For $m = 2, 4, 6, 8, 10$ the coordinate i has $E_i/\mathrm{Inf}_i =$ 2.672, 5.224, 7.387, 9.429, 11.439 and $\min_\pi L_i^\pi/\mathrm{Inf}_i =$ 2.950, 5.351, 7.426, 9.439, 11.442, while every coordinate j of y has $\min_\pi L_j^\pi/\mathrm{Inf}_j =$ 0.891, 1.699, 1.929, 1.983, 1.996. So E_i/Inf_i grows about linearly in m at i while every other coordinate stays below 2, and the choice of the coordinate in Conjecture 5 cannot be dropped.

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more on the details behind the Brain and Math Brain, see the Math Brain paper [17], the Math Brain repository [16] and the Brain framework repository [18].

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, a project on the use of AI systems in research mathematics [18, 16]. In November 2026 a Claude Fable agent (Anthropic), connected to the Brain and working alongside Codex agents running OpenAI's GPT-6.1 on the same problem, found the witness, its martingale form, the two-sequence condition and its resampling form, the composition formula, the private-AND bound and the windows. Claude Opus 5.5 (Anthropic) rederived each proof, wrote the numerical checks, prepared the manuscript and audited it. A further Claude agent found the bound of Remark 3 and checked it in Lean 4 with Mathlib (file in the code repository). The author directed the work and is responsible for the content of the paper.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-filtration-witness`, archived on Zenodo with doi:10.5281/zenodo.24125388.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [16].

Appendix A. Notation

Symbols are listed roughly in the order in which they first appear, and the last column gives that place. The letters a and b are the arguments of Φ and ψ and also name the sums $a(R)$ and $b(R)$; the letter l is an index in several proofs.

Symbol	Meaning	First appears
f, n	a Boolean function $\{-1,1\}^n \to \{-1,1\}$, and its number of variables	Section 1
x, x_j	a point of $\{-1,1\}^n$, and its j-th coordinate	Section 1
i, j	coordinates; i is the coordinate under study	Section 1
S, $\hat{f}(S)$, χ_S	a set of coordinates, the Fourier coefficient of f at S, and the character $\prod_{j\in S} x_j$	Section 1
U	the spectral sample, a random set with $\Pr[U=S]=\hat{f}(S)^2$	Section 1
$H(\cdot)$, $H(f)$	Shannon entropy in bits, and the Fourier entropy $H(U)$	Section 1

Symbol	Meaning	First appears
$\mathbb{E}$	expectation; over a uniform point x unless the variable is U	Section 1
$\mathrm{Inf}_i(f)$, $I(f)$	the influence $\Pr[i \in U]$ of the coordinate i, and the total influence $\sum_i \mathrm{Inf}_i(f)$	Section 1
C	the constant in the FEI conjecture $H(f) \le C\, I(f)$	Section 1
$\mathrm{Var}(f)$	the variance of f	Section 1
$E_i(f)$	the erasure entropy $H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$	Section 1
$\mathbf{1}[\cdot]$	the indicator of an event	Section 1
$J_i(f)$	the Jensen–Shannon term in the drop of the entropy from f to its restrictions at $x_i = \pm 1$	Section 1
c	an absolute constant, in (1) and in Conjecture 5	Section 1
π	an order of the coordinates other than i	Section 1
$L_i^\pi(f)$	the witness $\sum_k \Phi(A_k, B_k) = H(\mathbf{1}[i \in U] \mid \mathrm{lab}(U \setminus \{i\}))$	Section 1
D_j	the derivative in the direction j, $D_j u = \frac{1}{2}(u\|_{x_j=1} - u\|_{x_j=-1})$	Section 1
r	the length of a window, the number of last revealed positions that are recorded	Section 1
g, h	the average $\frac{1}{2}(f\|_{x_i=1} + f\|_{x_i=-1})$ and the derivative $D_i f$, functions of the coordinates other than i	Section 2
$\hat{g}(T)$, $\hat{h}(T)$	the Fourier coefficients of g and h at T	Section 2
δ	$\mathrm{Inf}_i(f) = \mathbb{E}h^2$; in Lemma 2, $\sum_k B_k$	Section 2
T	a subset of $[n] \setminus \{i\}$	Section 2
$p(T)$, $q(T)$	$\hat{f}(T)^2$ and $\hat{f}(T \cup \{i\})^2$	Section 2
$\Phi(a, b)$	$\eta(a) + \eta(b) - \eta(a + b)$, the entropy of telling a mass a from a mass b	Section 2
$\eta(x)$, h_2	$-x \log_2 x$, and the binary entropy function	Section 2
j_k, m	the k-th coordinate of the order π, and $m = n - 1$	Section 2
k	a step of the order, $0 \le k \le m$	Section 2
S_k	$\{j_1, \ldots, j_k\}$, the coordinates revealed after step k	Section 2
$\mathrm{lab}(T)$	the label of T, the largest k with $j_k \in T$, and 0 for $T = \emptyset$	Section 2
A_k, B_k	the p-mass and the q-mass of the sets with label k; in Lemma 2, any nonnegative numbers	Section 2
$\mathcal{F}_k$	the σ-algebra generated by $x_{j_1}, \ldots, x_{j_k}$	Theorem 1
R, R_*	a set of coordinates other than i, and the full set $[n] \setminus \{i\}$	Theorem 1
$a(R)$, $b(R)$	$\sum_{T \subseteq R} p(T)$ and $\sum_{T \subseteq R} q(T)$	Theorem 1
$\lambda(R)$	the value of the shortest-path recursion at R; $\lambda(R_*) = \min_\pi L_i^\pi$	Theorem 1
M, K	the constant in the product condition, and a cutoff $0 \le K \le m$	Lemma 2
$\psi(a, b)$	$b \log_2(1 + a/b)$	proof of Lemma 2
G, A_G	a set of indices, and $\sum_{k \in G} A_k$	proof of Lemma 2
β_k	$\delta^{-1} \sum_{l \le k} B_l$	proof of Lemma 2
t, G_t, k_t	a dyadic level, the indices k with $2^{-t-1} < \beta_k \le 2^{-t}$, and the least of them	proof of Lemma 2

Symbol	Meaning	First appears
l	an index: of a sum, of a block of a coarser chain, or of an element of Y	proof of Lemma 2
κ, τ_l	$1+2M$, and $\sum_{k\geq l} A_k$	Remark 3
x^R	x with the coordinates in R replaced by independent uniform bits	Section 3
$W(R)$, $V(R)$	$\mathbb{E}\, h(x)h(x^R)$, and $\mathbb{E}g^2 - \mathbb{E}\, g(x)g(x^R)$, the influence of the block R on g	Section 3
φ	a real function of the coordinates other than i, in Parseval's identity	Section 3
R_k	$R_* \setminus S_k$, the coordinates not yet revealed after step k	Theorem 4
R'_l	the sets of a coarser chain that the order refines	Section 3
$\mathcal{D}$, $\mathcal{L}$, $\mathcal{H}$	the sets $\{h \neq 0\}$, $\{g=-1\}$ and $\{g=1\}$	Section 3
s, w	the number and the width of the disjoint ANDs of a tribes function	Section 3
Q, P_0	the other $w-1$ variables of the AND that contains i, and the probability that no other AND is satisfied	Section 3
X, Z, z	the disjoint sets of coordinates of the inner and the outer function, and a point of $\{-1,1\}^Z$	Proposition 6
γ, F	the inner Boolean function on $\{-1,1\}^X$, and the outer one on $\{-1,1\}\times\{-1,1\}^Z$	Proposition 6
u, F_0, F_1	the first argument of F, and the parts in $F(u,z) = F_0(z) + uF_1(z)$	Proposition 6
$\hat{\gamma}$, $\hat{F}_0$, $\hat{F}_1$	the Fourier coefficients of γ, F_0 and F_1	Proposition 6
T_1, T_2	subsets of X and of Z	proof of Proposition 6
Y, d	the set of coordinates that enter only through the private AND, and its size	Proposition 7
ω	the private AND, the indicator that $x_i = 1$ and $x_j = 1$ for every $j \in Y$	Proposition 7
v, f_0, f_1	the remaining coordinates, and the functions with $f = f_0(v) + f_1(v)\,\omega$	Proposition 7
T_0, p_{T_0}	a subset of v, and $4^{-(d+1)}\hat{f}_1(T_0)^2$	proof of Proposition 7
μ, ν	the p-mass and the q-mass of the atoms of the sets $T = T_0$	proof of Proposition 7
e	the base of the natural logarithm	Proposition 7
$\mathrm{lab}_r(T)$, $L_i^{\pi,r}(f)$	the last r positions of π met by T, and $H(\mathbf{1}[i\in U] \mid \mathrm{lab}_r(U\setminus\{i\}))$	Section 6
$\mathcal{K}$, $D_{\mathcal{K}}$	a set of coordinates other than i, and the product of the D_j over $j \in \mathcal{K}$	Section 6
AND_n, MAJ_n, $\mathrm{TRIBES}_{w\times s}$	the AND and the majority of n variables, and the OR of s disjoint ANDs of width w	Section 7

Symbol	Meaning	First appears
y, $N(y)$, $\mathrm{IP}(y)$	a point of $\{-1,1\}^m$, the number of $j \le m/2$ with $y_j = y_{j+m/2} = -1$, and the inner product function $(-1)^{N(y)}$	Section 7

References

[1] E. Friedgut, G. Kalai, Every monotone graph property has a sharp threshold, Proc. Amer. Math. Soc. 124 (1996), 2993–3002.

[2] S. Arunachalam, S. Chakraborty, M. Koucký, N. Saurabh, R. de Wolf, Improved bounds on Fourier entropy and min-entropy, ACM Trans. Comput. Theory 13 (2021), no. 4, 1–40, doi:10.1145/3470860; STACS 2020, arXiv:1809.09819.

[3] S. Chakraborty, R. Kulkarni, S. V. Lokam, N. Saurabh, Upper bounds on Fourier entropy, Theoret. Comput. Sci. 654 (2016), 92–112.

[4] M. J. González, P. MacManus, M. C. Pereyra, Further evidence towards the Fourier entropy-influence conjecture, arXiv:2606.00246, 2026.

[5] X. Han, A new bound for the Fourier-entropy-influence conjecture, Combinatorica 45 (2025), no. 1, Art. 4, doi:10.1007/s00493-024-00133-z; arXiv:2312.08271.

[6] E. Kelman, G. Kindler, N. Lifshitz, D. Minzer, M. Safra, Towards a proof of the Fourier-entropy conjecture?, Geom. Funct. Anal. 30 (2020), 1097–1138.

[7] P. Li, G. Han, Strengthening Han's Fourier entropy-influence inequality via an information-theoretic proof, arXiv:2512.03117, 2025.

[8] R. O'Donnell, L.-Y. Tan, A composition theorem for the Fourier entropy-influence conjecture, ICALP 2013, Lecture Notes in Comput. Sci. 7965, 780–791, doi:10.1007/978-3-642-39206-1_66.

[9] R. O'Donnell, J. Wright, Y. Zhou, The Fourier entropy-influence conjecture for certain classes of Boolean functions, ICALP 2011, Lecture Notes in Comput. Sci. 6755, 330–341.

[10] Y. Shmalo, The one-coordinate step in the Fourier entropy-influence conjecture, `https://github.com/yspennstate/fourier-entropy-influence-one-coordinate`, doi:10.5281/zenodo.24125404, 2026.

[11] Y. Shmalo, Good coordinates in the Fourier entropy-influence conjecture, `https://github.com/yspennstate/fei-good-coordinates`, doi:10.5281/zenodo.24125398, 2026.

[12] Y. Shmalo, A variance form of Han's bound on the Fourier entropy, `https://github.com/yspennstate/fei-variance-han-bound`, doi:10.5281/zenodo.24125394, 2026.

[13] Y. Shmalo, Fourier entropy of threshold functions and their compositions, arXiv:2610.16069, 2026.

[14] S. Verdú, T. Weissman, The information lost in erasures, IEEE Trans. Inform. Theory 54 (2008), 5030–5058.

[15] A. Wan, J. Wright, C. Wu, Decision trees, protocols, and the Fourier entropy-influence conjecture, ITCS 2014, arXiv:1312.3003.

[16] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[17] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`, archived on Zenodo with doi:10.5281/zenodo.24150837.

[18] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.

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